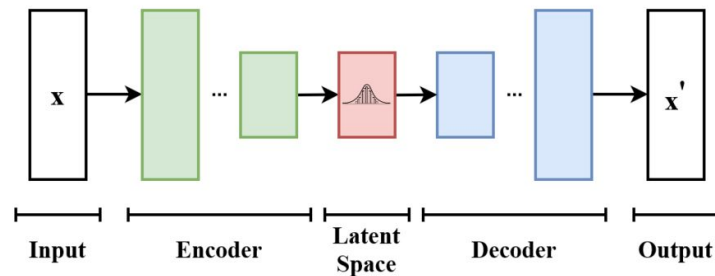
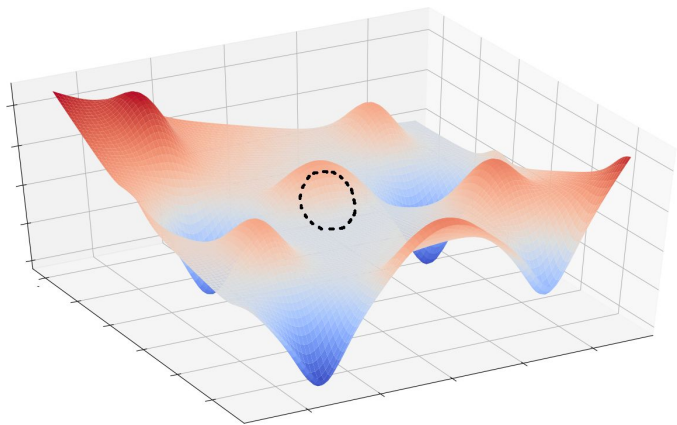


LatentCraft

Property optimization via latent editing

theoretical background

Latent space is a compressed, continuous representation of objects that still retains key information about their manifestations and properties.

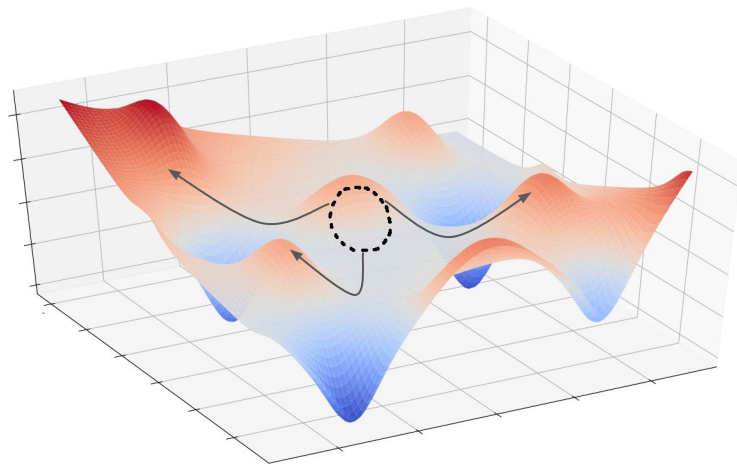


Due to the smoothness of space, **directed movement** in it leads to predictable changes in the properties of decoded objects.

key concept

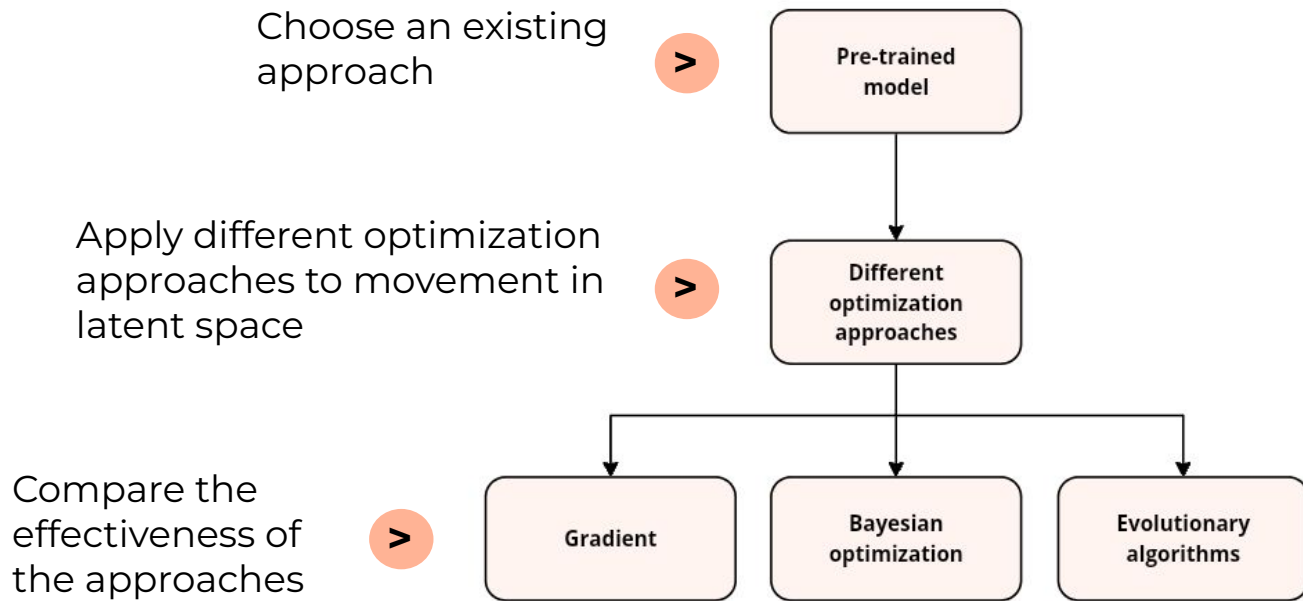
- 1 Build a continuous latent space that compresses discrete chemical structures
- 2 Pick a seed molecule and map it to a latent vector
- 3 Navigate the latent space toward regions with improved target properties
- 4 Apply uncertainty control to stay in valid regions

Result: novel molecules — optimized and structurally close to the seed



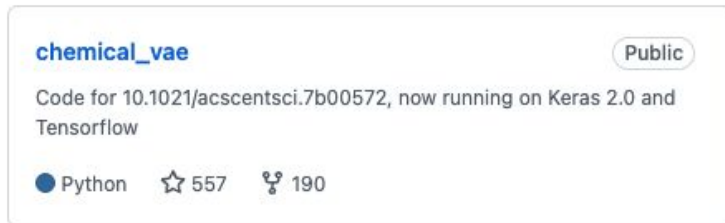
project's goal: perform a comparative analysis of architectures and optimization approaches for the generation of new molecules using motion in latent space.

experimental design



model architectures: chemical VAE

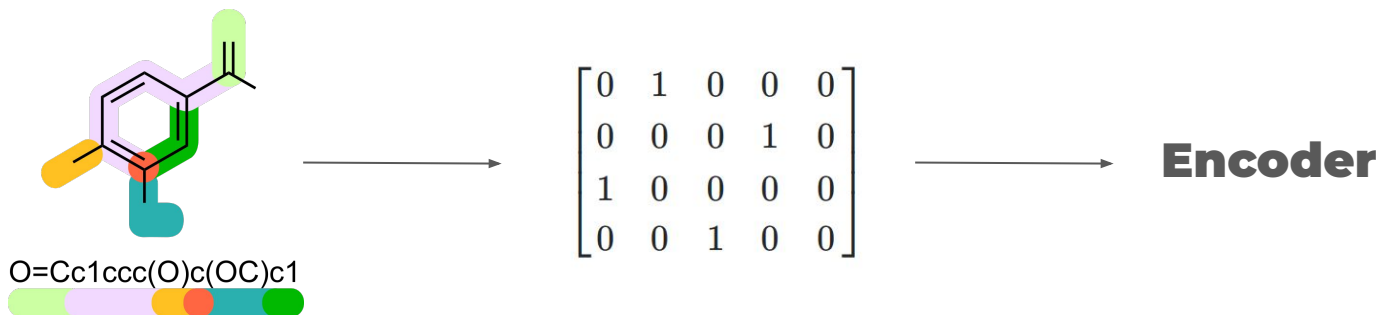
Authors: Gómez-Bombarelli R. et al. 2018



Architecture & Method:

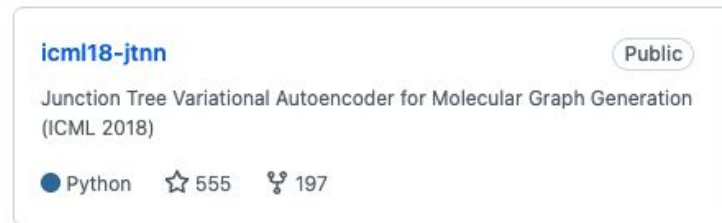
Encodes molecular SMILES into a continuous latent vector using a variational autoencoder jointly trained with property predictors to shape the latent space.

Code Repository: github.com/aspuru-guzik-group/chemical_vae



model architectures: JT-VAE

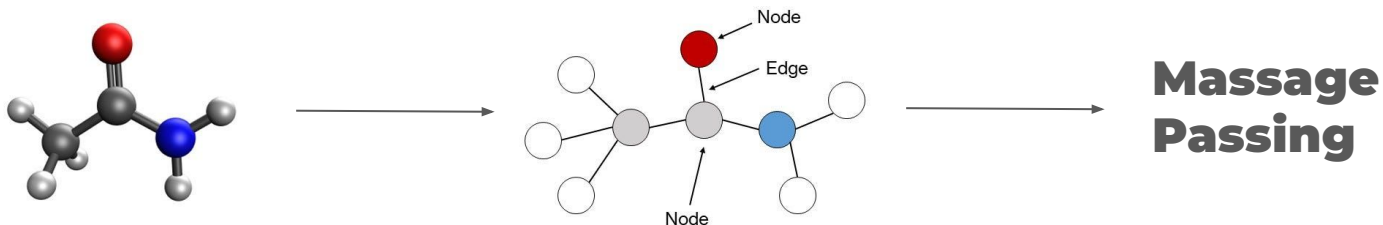
Authors: Jin W. et al. 2018 (ICML)



Architecture & Method:

The architecture employs a dual-latent representation that first generates a tree-structured scaffold of chemical substructures and then assembles it into a full molecular graph via message passing networks.

Code Repository: github.com/wengong-jin/icml18-jtnn



optimization methods

Gradient ascent

- Encode the seed molecule
- Move the latent vector using a pre-trained predictor.

Very fast, deterministic, and easy to implement.

Bayesian Optimization

- Build a probabilistic surrogate model
- Balance exploration of new regions with exploitation of known good ones.

Sample-efficient, handles noise well, and provides uncertainty estimates.

CMA-ES

- A evolutionary strategy
- Adapts a multivariate Gaussian distribution over the latent space to shift the search toward promising areas.

Global search capability, rotation-invariant, and robust to non-smooth property landscapes.

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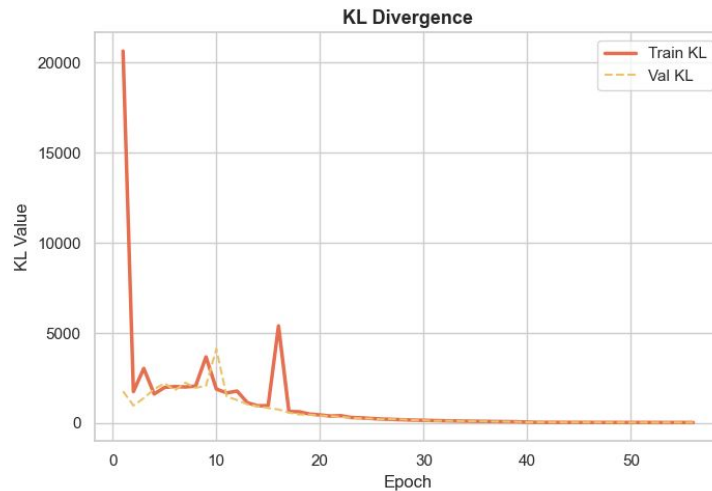
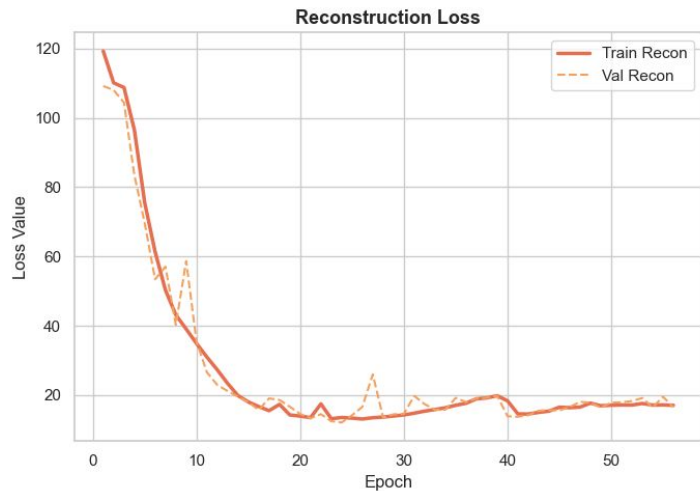
CMA-ES

- A evolutionary strategy
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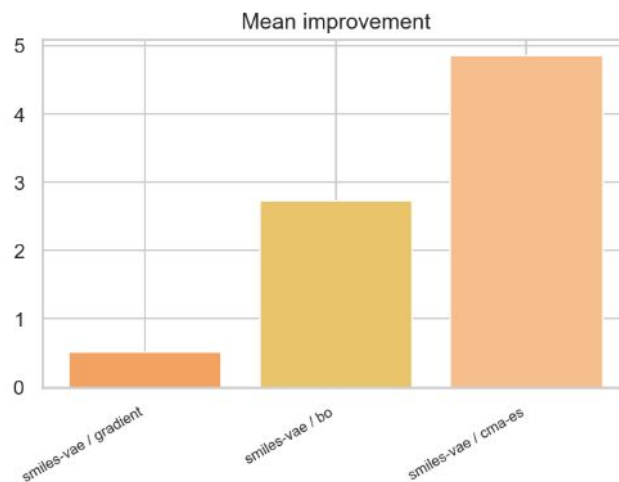
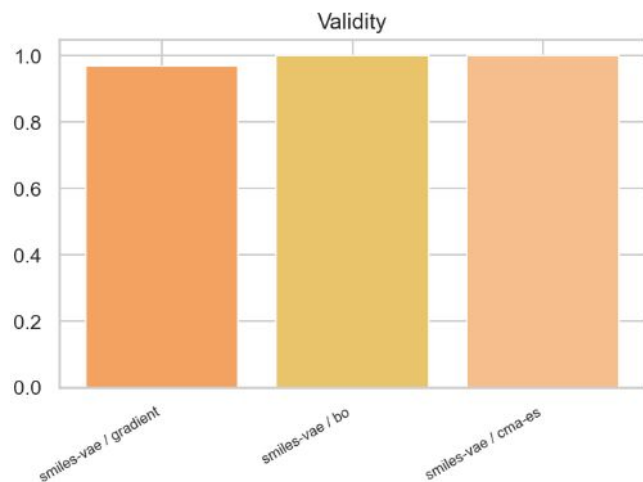
Global search capability, rotation-invariant, and robust to non-smooth property landscapes.

chemical VAE training

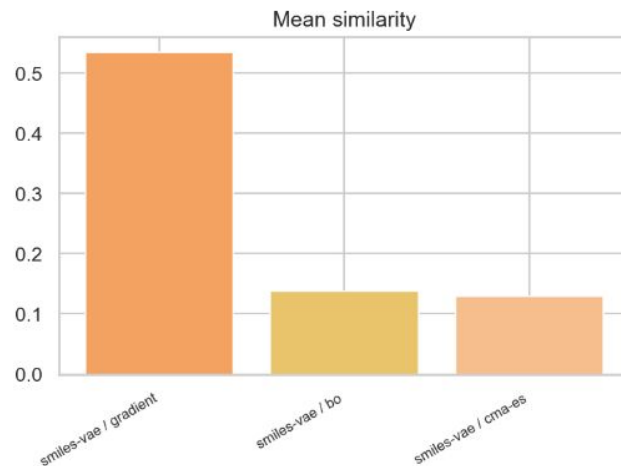
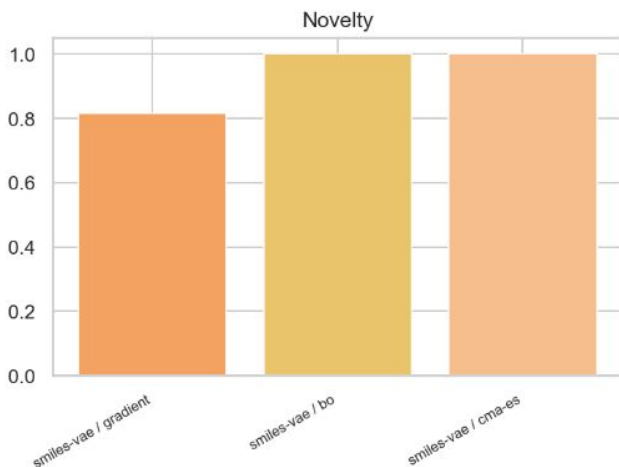
- 120 epochs + early stopping
- optimizer: Adam
- loss: $L = \beta \cdot \text{KL} + \text{Reconstruction} + \lambda \cdot \text{MSE}(\log P)$
- Annealing strategy: sigmoid function



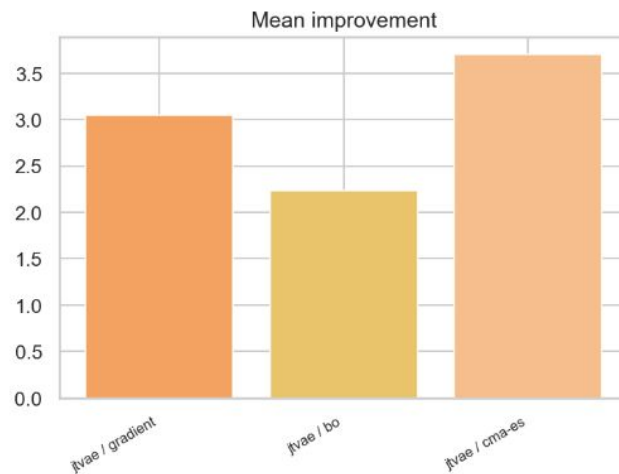
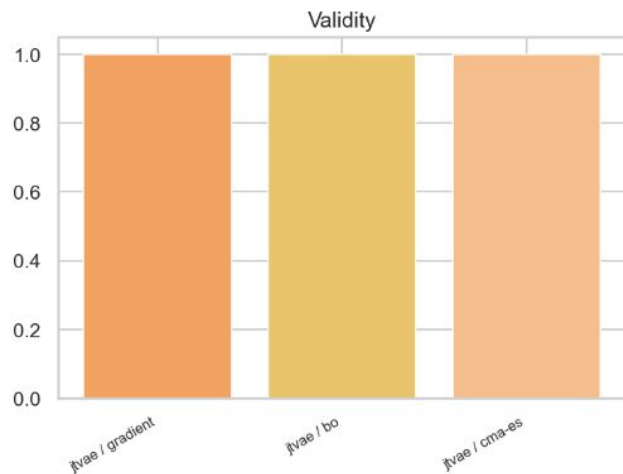
chemical VAE optimization results



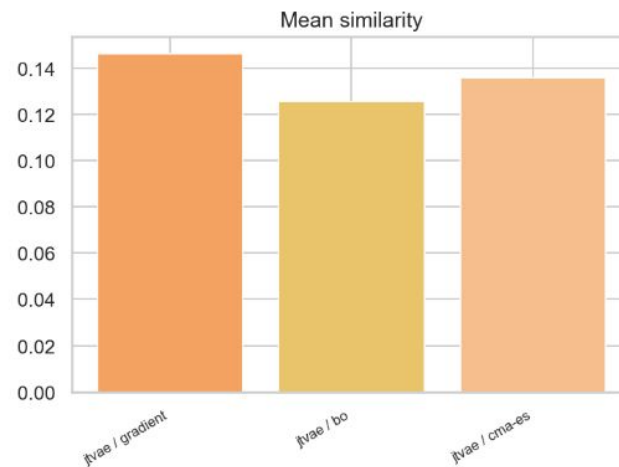
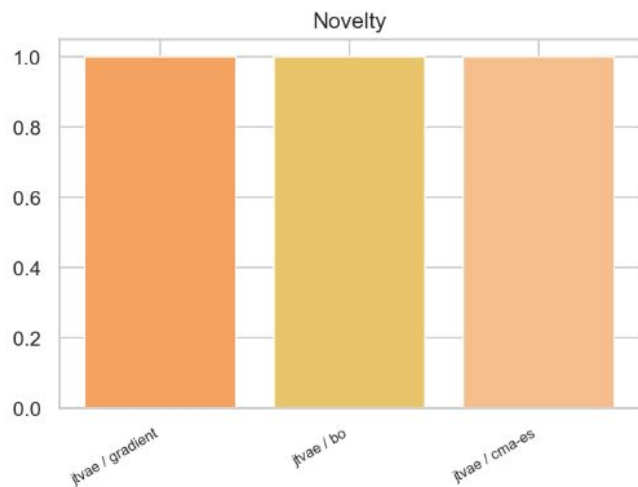
chemical VAE optimization results



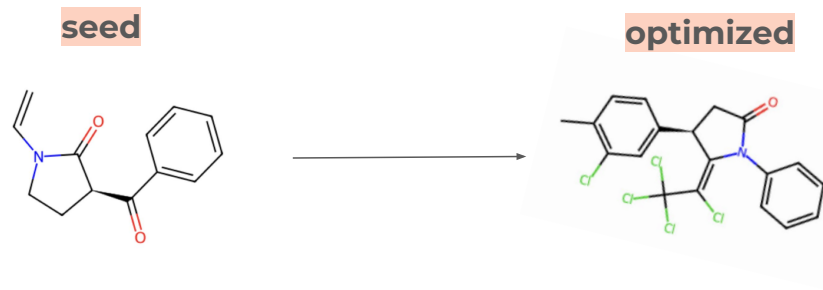
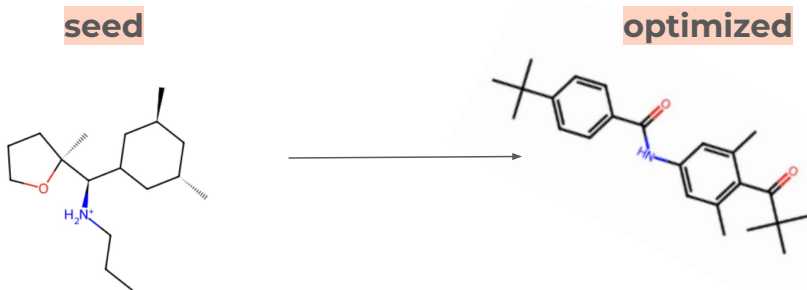
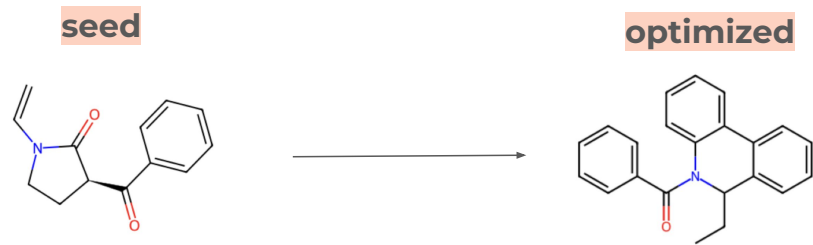
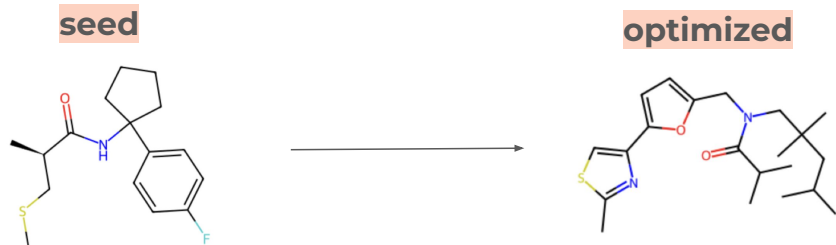
JT-VAE optimization results



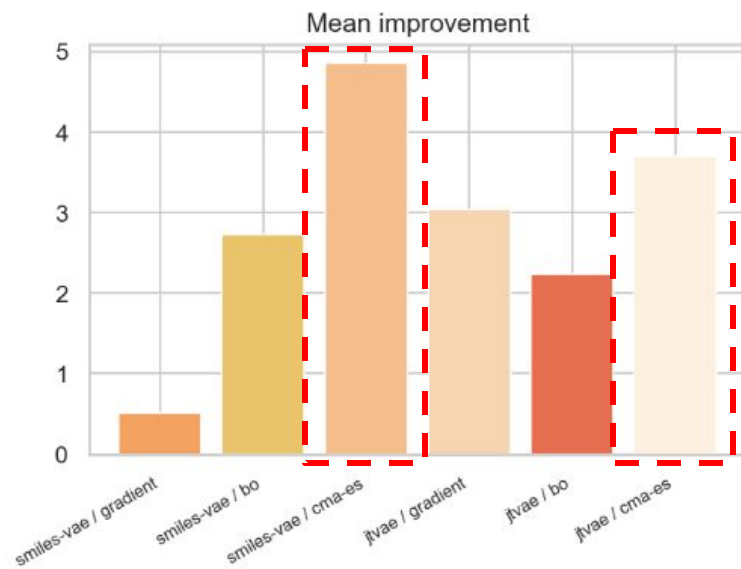
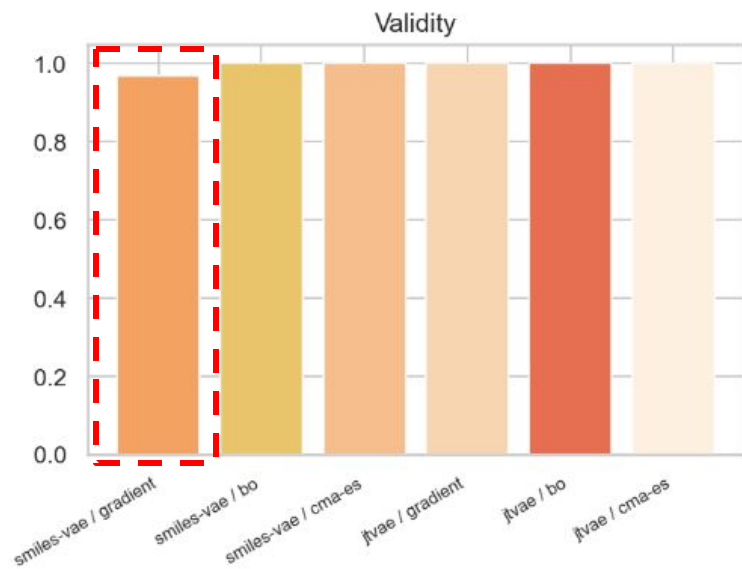
JT-VAE optimization results



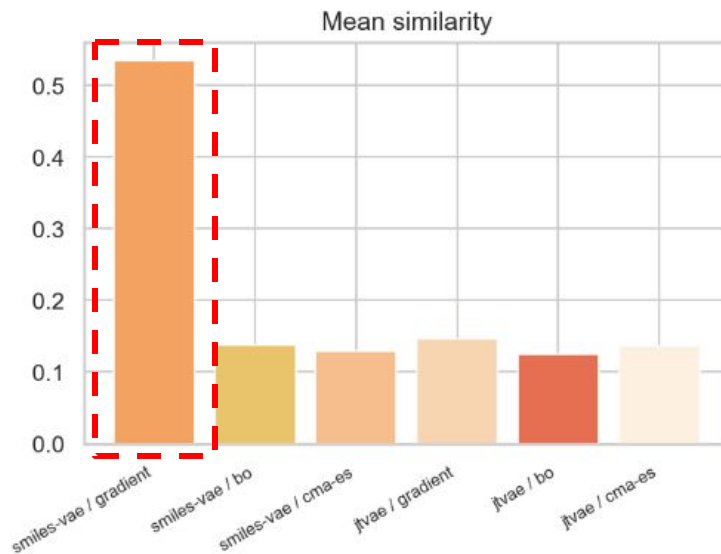
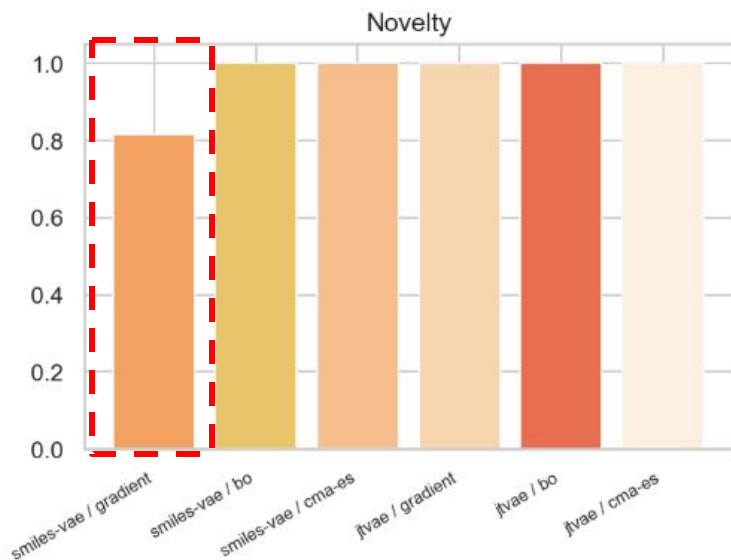
examples of generated molecules



comparison



comparison



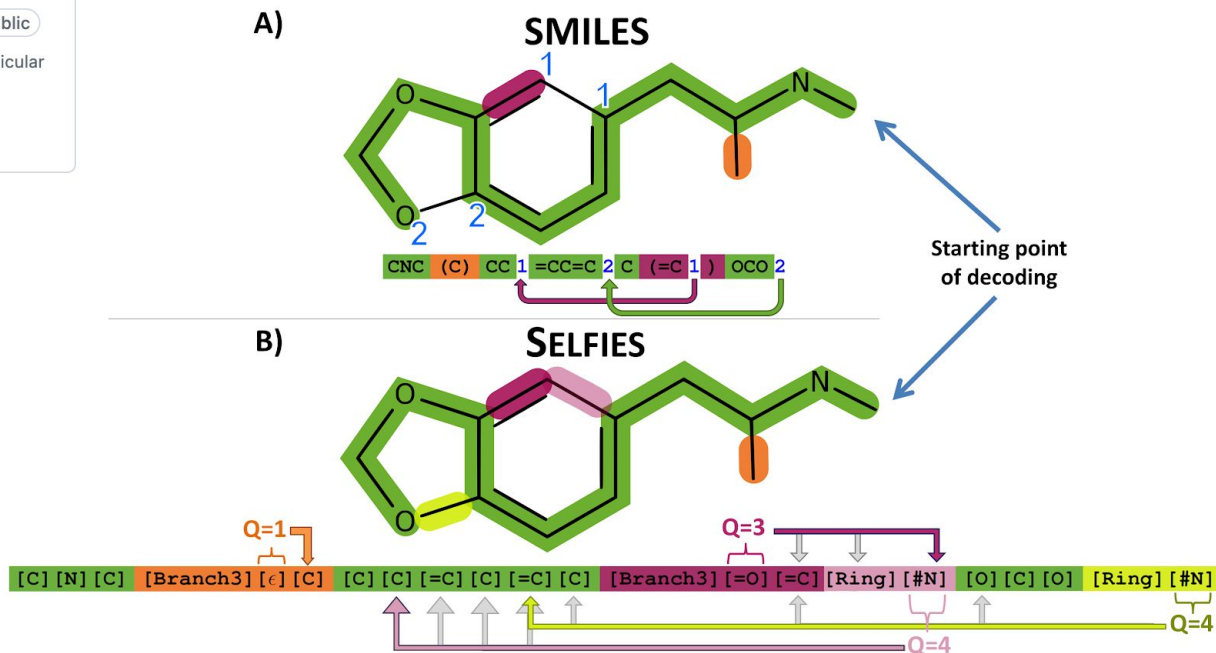
*experiments: SELFIES framework

selfies

Public

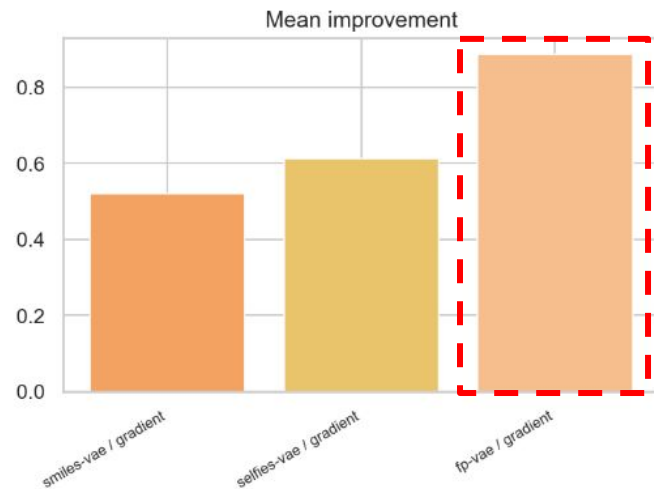
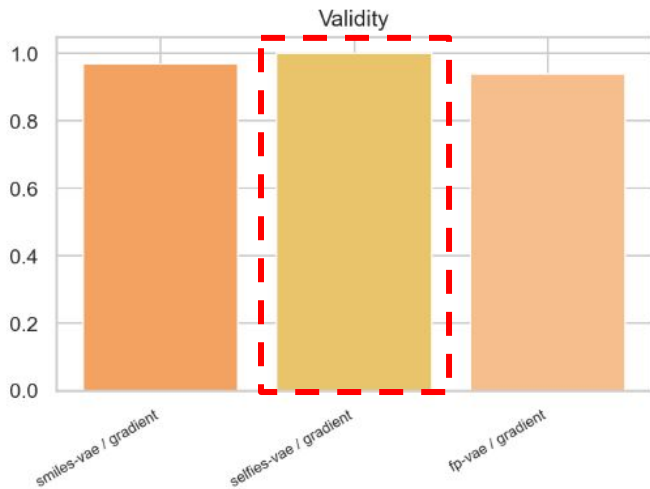
Robust representation of semantically constrained graphs, in particular for molecules in chemistry

Python 843 146



***experiments: advanced encoding**

*experiments: results



discussion

references

supplementary 1

supplementary 2

supplementary 3